# Muneeb Arshad

Potsdam, NY

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## EDUCATION

**CLARKSON UNIVERSITY** Potsdam, NY, USA
*PhD Candidate, Electrical and Computer Engineering* January 2022 – December 2026

- Coursework: Deep Learning, Space Robotics, Orbital Mechanics and Dynamics, Advanced Computer Aided Design, Software System Architecture
- Research: Soft Gripper-Based Space Debris Capture Using Deep Reinforcement Learning

**CLARKSON UNIVERSITY** Potsdam, NY, USA
*Master of Science in Electrical and Computer Engineering* January 2022 – April 2025

**NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY** Islamabad, Pakistan
*Master of Science in Electrical Engineering (Control System)* September 2019 – December 2021

- Coursework: Advanced Digital Signal Processing, Machine Learning, Linear Control Systems, Stochastic Systems, Advanced Power Electronics, Image Processing for Intelligent Systems, Artificial Intelligence, Mobile Robotics
- Thesis: Analysis and Classification of EEG Signals

**COMSATS UNIVERSITY** Islamabad, Pakistan
*Bachelor of Science in Electrical and Computer Engineering* September 2015 – April 2019

- Final Year Project: Design of Auto tracker for Solar Panels

## EXPERIENCE

**YORK UNIVERSITY** Toronto, Canada
***International Visiting Research Trainee*** January 2024 – January 2025

- Conducted research on collision avoidance and disturbance minimization for a free-floating space manipulator using deep reinforcement learning-based control method

**MAANZ-AI** Islamabad, Pakistan
***Associate C++ Developer*** January 2021 – December 2021

- Contributed to the development of a simulator for self-driving cars

**ENGINEERING DEVELOPMENT BOARD** Islamabad, Pakistan
***Intern*** March 2020 – December 2020

- Contributed to the development of a policy for power equipment and the energy sector

**NATIONAL INSTITUTE OF ELECTRONICS** Islamabad, Pakistan
***Intern*** July 2018 – August 2018

- Contributed to the development and testing of custom PCBs

## RESEARCH INTERESTS

Artificial intelligence, reinforcement learning, advanced control, soft robotics, space robotics, and mixed reality for enabling autonomous on-orbit operations. Key areas include space debris capture, in-orbit assembly, and soft robotic manipulation for space applications, with extensions to terrestrial soft robotics.

## RESEARCH PUBLICATIONS

- M. Arshad, M.C.F. Bazzocchi, F. Hussain, "Emerging Strategies in Close Proximity Operations for Space Debris: A Review" Acta Astronautica, Vol. 228, pp. 996-1022, 2025.
- M. Arshad, M.C.F. Bazzocchi, "Collision Avoidance and Disturbance Minimization through Deep Reinforcement Learning Control of a Free-Floating Space Manipulator," 75th International Astronautical Congress (IAC), Milan, Italy, Oct. 14-18, 2024.
- M. Arshad, M. C. F. Bazzocchi, "Smooth and Safe Pre-Capture Control of a Free-Floating Space Manipulator Using Deep Reinforcement Learning Control," under review.
- M. Arshad, M. C. F. Bazzocchi, "Deep Reinforcement Learning for Stable Grasping of a Non-Cooperative Space Target Using a Soft Robotic End-Effector," in preparation.

- M. Arshad, M. C. F. Bazzocchi, “Deep Reinforcement Learning Control for detumbling a Non-Cooperative Space Target,” in preparation.

## PEER REVIEWER FOR JOURNALS

- Reviewed 16 research manuscripts as a peer reviewer for leading Elsevier space journals, including Aerospace Science and Technology, Acta Astronautica, and Advances in Space Research, providing detailed technical feedback to inform publication decisions.

## PRESENTATIONS

- Muneeb Arshad, *“Collision Avoidance and Disturbance Minimization through Deep Reinforcement Learning Control of a Free-Floating Space Manipulator”*, 75th International Astronautical Congress (IAC), Milan, Italy, Oct. 14-18, 2024.

## HONORS AND AWARDS

- Tuition fellowship – Clarkson University
- US-Pakistan Knowledge Corridor Scholarship

## CERTIFICATIONS

- Robotics in Space – International Space Alliance (ISA)
- Artificial Intelligence - Presidential Initiative for Artificial Intelligence and Computing (PIAIC)
- Level 2: Master Certification – STK
- Level 1: STK
- AI for Everyone – Coursera

## SOFTWARE SKILLS

- MATLAB, Xilinx, ORCAD, Proteus, Atmel Studio, Visual Studio, SolidWorks, ANSYS, CREO, AutoCAD, MeshLab, 3D Zephyr, Fusion 360

## PROGRAMMING SKILLS

- Python, C++, C, Assembly language, JAVA, Scala

## OPERATING SYSTEMS

- Windows, Linux

## EXTRA CURRICULAR ACTIVITIES

- Participated in Inter-Departmental Quiz Competition, COMSATS University Islamabad
- Participated in Robotics Competition, COMSATS University Islamabad
- Member, Pakistani Student Association, Clarkson University

## WORKSHOPS

- National Workshop on Application of Robotics Technology for Industrial Development - NUST
- Arduino - COMSATS University
- PCB fabrication - COMSATS University
- National police summit and innovation expo - Islamabad

## ADDITIONAL INFORMATION

- Languages: Urdu (Native), English (Advanced)
- Interests: Horse Ridding, Swimming, Cricket, Volleyball